

## LESSON 2.2: File Layers

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### What this lesson is about

This lesson provides an overview of supported file layers in Sterling File Gateway, will also.

### What you should be able to do

- Explain the importance of file layers in Sterling File Gateway.
- Discuss to add custom file layer in Sterling File Gateway.
- Explain to extract compressed file layer.
- Create a routing channel template and routing channel for layered file.

### How you will check your progress

The progress of the course is analyzed based on the successful application of the hands-on exercises.

### Additional References

#### **Documentation**

- Sterling File Gateway custom file layers:  
[http://www.ibm.com/support/knowledgecenter/SS4TGX\\_2.2.0/com.ibm.help.sfg\\_sys\\_admin.doc/SFGAboutCustomFileLayers.html?lang=en](http://www.ibm.com/support/knowledgecenter/SS4TGX_2.2.0/com.ibm.help.sfg_sys_admin.doc/SFGAboutCustomFileLayers.html?lang=en)
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## File layers

### Introduction

In Sterling File Gateway, File Layers are used to describe the layers or wrappings around the innermost file. Sterling File Gateway uses the file layer description to transform the producer file into a format that meets the consumer's business requirements. This file brokering capability enables significant file transfer flexibility.

The routing channel template establishes descriptions of the files that are expected from the producer through the producer file structure and the files consumer require through the consumer file structure.

It is possible to send and receive files nested within other files. For example, a compressed file that contains multiple JPEG files. The routing channel template allows expression of file descriptions in layers, with each layer representing a type of file.

### Standard file layers

The built in standard file layers in Sterling File Gateway are shown in the table:

| Layer         | Type    | Description                                                                                      | Contains other files |
|---------------|---------|--------------------------------------------------------------------------------------------------|----------------------|
| Container     | ZIP     | a file with a *.zip extension                                                                    | Yes                  |
|               | GZIP    | a file with a *.gzip extension                                                                   | Yes                  |
|               | PGP     | a file with a *.pgp extension or a *.asc extension (to represent PGP with ASCII armor turned on) | Yes                  |
| Non-Container | TEXT    | a file with a *.txt extension                                                                    | No                   |
|               | UNKNOWN | any type of file                                                                                 | No                   |

Container type layer: Container type layer contains other container or non-container layers, such as a PGP layer can contain ZIP layer and ZIP layer can contain a text layer. The standard container layers are Zip, GZip, PGP.

Non-container type layer: Non-container type layer does not contain nested layers. It is also called as primitive. The standard non-containers layers are Text and Unknown.

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## File layers

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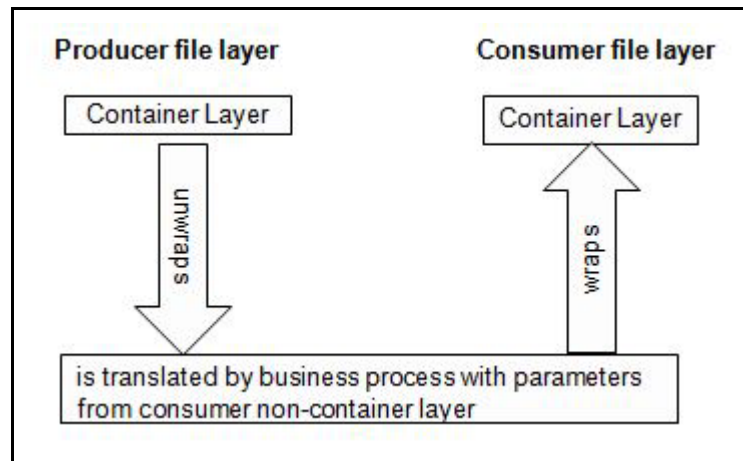
### Introduction

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#### Processing file layer

The screen shot represents the relationship between the file layers that Sterling File Gateway performs between the layer when the producer and the consumer configure the file layers in routing channel template. When the producer transfers the file in container layer format, the consumer will identify the producer and uses the business process to translate to the specific parameters and wraps into the consumer file structure.

Sequence of the layer mappings are producer *container* > producer *non-container* > consumer *non-container* > consumer *container*..



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## File layers

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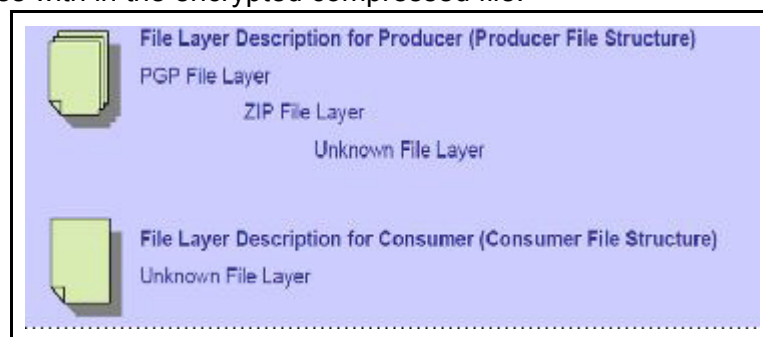
### Introduction

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#### File layer description

Sterling File Gateway uses the file layer descriptions to transform the producer file into a format that meets the consumer's business requirements. If the producer file does not fit the description in the routing channel template, then the file is not routed.

For example, Consider a file transfer between a vehicle manufacturer that sends encrypted compressed files that contain multiple CAD drawings to a supplier who uses the CAD drawings to design navigation systems for several car models. The supplier is only interested in the individual CAD files with in the encrypted compressed file.



The innermost layer of a producer nested file description maps to a file structure for the consumer. There is no mapping for the outer layers. The table here describes supported mapping for innermost layers of a producer file and consumer file. Using the CAD drawing example, the Unknown producer file maps to an Unknown consumer file.

|                         | Consumer File > TEXT | Consumer File > UNKNOWN |
|-------------------------|----------------------|-------------------------|
| Producer File > TEXT    | Good                 | Good                    |
| Producer File > UNKNOWN | Bad                  | Good                    |

## Custom file layer

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### Introduction

Following are the attributes of custom file layer:

- The custom file layers can be of container or non-container type for producer or consumer file layers. You can create a maximum of 100 customer layer types. The customer container layer must be configured with a business process that will perform the translation from one format to the other.
- The name of the custom file layer types must begin with “FGC\_”. FGC indicates table entries for the LAYER\_TYPE column in the FG\_P\_FLR\_TYPE and FG\_C\_FLR\_TYPE tables. The maximum length for LAYER\_TYPE is 10 characters including FGC\_.
- The business process performs translation, container layer transformation, container layer creation and consumer identification.
- Sterling File Gateway invokes custom file layer business process synchronously.
- Example for customer file layer is translation with any transformation of a file.

For more details about custom file layers, refer

[http://www.ibm.com/support/knowledgecenter/SS4TGX\\_2.2.0/com.ibm.help.sfg\\_sys\\_admin.doc/SFGAboutCustomFileLayers.html?lang=en](http://www.ibm.com/support/knowledgecenter/SS4TGX_2.2.0/com.ibm.help.sfg_sys_admin.doc/SFGAboutCustomFileLayers.html?lang=en)

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## Exercise 2.2.1: Unwrapping an Inbound Compressed file into a Text File

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### Introduction

The exercise demonstrates how to create a routing channel template and a routing channel that extracts files from a compressed file and routes the individual files to the consumer's mailbox.

The exercise is a four step process.

1. Create a compressed file with two or more text files. The name of the text files must be in the format: ConsumerName\_FileName.txt. The name of the compressed file must be in the format: date.zip.
2. Create a Dynamic Inbound Routing Channel Template.
3. Create a Routing Channel.
4. Log in to myfilegateway with the ID of partner and upload the compressed file.

### Instructions

#### **Create a compressed file**

Complete the following steps to create a compressed file:

**Step 1:** From the windows image, access **C:\6F86G\zipexercise** directory.

**Step 2:** The following files are in zipexercise directory:

- consumername\_readme1.txt
- consumername\_readme2.txt
- consumername\_readme3.txt

**Step 3:** Rename the files to your company ID\_readme.txt as stated in the table:

| File name            | New file name     |
|----------------------|-------------------|
| consumer_readme1.txt | abcde_readme1.txt |
| consumer_readme2.txt | abcde_readme2.txt |
| consumer_readme3.txt | abcde_readme3txt  |

**Step 4:** Compress the files as 20162503.zip.

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## Exercise 2.2.1: Unwrapping an Inbound Compressed file into a Text File

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### Instructions

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#### Create a Routing Channel Template

Complete the following steps to create a routing channel template:

**Step 1:** Log in to filegateway by specifying **fg\_sysadmin** as user ID and **password** as password

**Step 2:** From the main menu, navigate to **Routes > Templates**.

**Step 3:** Click **+Create**.

**Step 4:** Enter Template Name: **Dynamic\_datetime\_inbound**.

**Step 5:** Enter Template Type: **Dynamic**, specify the following values:

| Parameter                         | Value                            |
|-----------------------------------|----------------------------------|
| What will determine the consumer? | Facts in producer file structure |

**Step 6:** From the **Special Characters** tab. Select **None** for the **Special Characters**.

**Step 7:** In the **Groups** tab, select **+Add** to add the **Producer Groups**.

**Step 8:** From the drop down list, select **All Partners**.

**Step 9:** Click **+Add** to add the **Consumers Groups**.

**Step 10:** From the drop down list, select **All Partners**.

**Step 11:** Accept the default values for **Provisioning Facts** tab.

**Step 12:** In the **Producer** tab, select **+Add** to add the **Producer**.

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## Exercise 2.2.1: Unwrapping an Inbound Compressed file into a Text File

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### Instructions

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**Step 13:** In the **Producer File Structure**, specify the following values:

| Parameter                                            | Value    |
|------------------------------------------------------|----------|
| Producer File Type:                                  | Zip      |
| File name pattern as regular expression:             | (.+).zip |
| File name pattern group fact names, comma delimited: | myDate   |

**Step 14:** Click **Next**.

**Step 15:** In **Producer File Structure**, specify the following values.

| Parameter                                            | Value                   |
|------------------------------------------------------|-------------------------|
| Producer File Type:                                  | Text                    |
| File name pattern as regular expression:             | (.+)_(.+).txt           |
| File name pattern group fact names, comma delimited: | ConsumerName,myFilename |

**Step 16:** Click **Save**.

**Step 17:** In the **Consumer** tab, click **+Add** to add a **Consumer**.

**Step 18:** Accept the default values for mailbox path.

**Step 19:** In the **Consumer File Structure**, specify the following values:

| Parameter          | Value              |
|--------------------|--------------------|
| Consumer File Type | Text               |
| File name format   | \${myFilename}.txt |

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## Exercise 2.2.1: Unwrapping an Inbound Compressed file into a Text File

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### Instructions

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**Step 20:** Click **Save** three times. The system displays **Routing Channel Template successfully created**. Click **OK**.

#### To create a routing channel

Complete the following steps to create a routing channel:

**Step 1:** Log in to filegateway by specifying **fg\_sysadmin** as user ID and **password** as password.

**Step 2:** From the main menu, navigate to **Routes > Channels**.

**Step 3:** Click **+Create**.

**Step 4:** In the **Create a New Routing Channel** window, specify the following values:

| Parameter                | Value                    |
|--------------------------|--------------------------|
| Routing Channel Template | Dynamic_datetime_inbound |
| Producer                 | partner                  |

**Step 5:** Click **Save**. The system displays **Successfully created Routing Channel**. Click **OK**.

#### Upload input data into myfilegateway

Complete the following steps to upload the file:

**Step 1:** Log in to **myfilegateway** by specifying **partner** as user ID and **password** as password.

**Step 2:** Click **Upload File** tab.

**Step 3:** In the Mailbox path section, select / [forward slash].

**Step 4:** In the File section, select **C:\6F86G\zip exercise\20162503.zip**.

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## Exercise 2.2.1: Unwrapping an Inbound Compressed file into a Text File

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### Instructions

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**Step 5:** Click **Send**. The file is routed successfully to `/abcde/inbox` as shown in the screen shot.



### Verify the files are transferred

**Step 1:** Log in to filegateway by specifying **fg\_sysadmin** as user ID and **password** as password.

**Step 2:** Navigate to **Routes > Activity**.

**Step 3:** Click **Find** to view the routed files as shown in the screen shot.



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## Exercise 2.2.1: Unwrapping an Inbound Compressed file into a Text File

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### Instructions

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Verify the file is routed in abcde partner interface to download the file.

**Step 1:** Log in to myfilegateway by specifying **abcde** as user ID and **password** as password.

**Step 2:** Select the **File Activity** tab.

**Step 3:** Click **Find** button in Search Criteria panel to view the arrived files in abcde's inbox.

**Step 4:** Click **Logout**.

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## Lesson review

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### What you have been able to do

- Explain the importance of file layers in Sterling File Gateway.
  - Discuss to add custom file layer in Sterling File Gateway.
  - Explain to extract compressed file layer.
  - Create a routing channel template and routing channel for layered file.
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